Chapter 1: Industrial Overheating and Thermal Degradation Diagnosis

Overheating is the leading catalyst for mechanical fatigue and lubrication film breakdown.

1.1 Heat Dissipation Failure (HDF) Diagnostic Rules:
- Rule 1: Temperature Delta Drop. Under steady load (Speed > 1400 rpm, Torque > 35 Nm), the difference between Process Temperature (T_proc) and Ambient Air Temperature (T_air) must remain strictly above 8.6 K.
- If (T_proc - T_air) < 8.6 K while rotational speed is below 1380 rpm, cooling flow insufficiency is confirmed.
- Common Causes:
  1. Coolant pump impeller wear or drive belt slippage.
  2. Bio-film or mineral scale accumulation in the internal heat exchanger tubes.
  3. Thermal bypass valve stuck in recirculation mode.

1.2 High-Pressure Compressor Thermal Divergence:
- In turbofans, an abnormal increase in s_3 (T30) while fuel flow ratio s_12 (phi) remains constant points to internal gas recirculation across eroded blade tips.
- If T30 rises above 1600 R accompanied by s_7 (P30) drop, shut down machine to prevent thermal surge.

Chapter 2: Root-Cause Fault Tree for Rotational & Bearing Anomalies

Vibration analysis provides the earliest warning of rotating component degradation.

Fault Tree 1: High Vibration (> 4.5 mm/s RMS):
- Predominant frequency 1x RPM: Unbalance or rotor mass asymmetry. Clean rotor, perform field balancing.
- Predominant frequency 2x RPM: Shaft misalignment or bent spindle. Verify dial indicator alignment across flexible coupling.
- High-frequency broadband noise (2 kHz - 10 kHz): Bearing raceway pitting, micro-spalling, or lubrication starvation.

Fault Tree 2: Rapid Tool Wear Failure (TWF) on CNC Spindles:
- Wear threshold exceeded: Tool wear minutes > 200 min.
- Combined metric: Torque (Nm) x Tool Wear (min) > 11,000.
- Action: Emergency tool insert replacement; inspect workpiece surface roughness and spindle collet clamping force.